Ajay Devidas Wanekar

M.Tech Energy Science and Technology - TA

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SUMMARY

M.Tech student at IIT Hyderabad with 2+ years of research experience in AI and Data Science, specializing in Computer Vision, Deep Learning, and Generative AI. Experienced in translating real-world problems into well-defined ML problems; building image-processing, detection, and segmentation pipelines (OpenCV, YOLO, Vision-Language Models); LLM and RAG systems; and classical ML (Random Forest, SVM, XGBoost) with rigorous statistical validation and reproducible, published work.

Degree	University/Institute	Year	CGPA/ Marks(%)
M.Tech Energy Science and Technology - TA	IIT Hyderabad	2027	6.8
B.E. Electrical Engineering	Savitribai Phule Pune University	2022	7.28
XII (Maharashtra State Board)	Vasantrao Naik Jr. College of Science and Arts, Cidco, Ch. Sambhajinagar, Maharashtra	2017	67.38%
X (Maharashtra State Board)	Maharashtra Public School, DTP, Ch. Sambhajinagar	2014	83.00%

WORK EXPERIENCE

Project Associate | LFOVIA Lab, IIT Hyderabad Sep 2024 – Jul 2025

Evaluated and fine-tuned Vision-Language Models (Florence-2 Base/Large, Llama 3.2 Vision, Qwen2-VL) for object detection, segmentation, and text-image understanding, and benchmarked them against task-specific deep-learning models (RetinaFace, DeepFace, Siamese networks) for detection, face analysis, and visual tracking using precision, recall, and IoU.

Junior Research Assistant | TIHAN, IIT Hyderabad: Nov 2023 – Jul 2024

Built real-time object detection and multi-object tracking with YOLO and DeepSORT, and curated near-infrared and sonar datasets with sensor-specific image preprocessing, annotation, data-quality optimization, and model-training pipelines for robust detection in complex environments, resulting in an IEEE conference publication.

Research Intern | WINET Lab, IIT Hyderabad: Jul 2023 – Nov 2023

Developed sonar-based underwater object-detection datasets with annotation, Python-based preprocessing, data augmentation, and quality checks to improve model robustness, published at IEEE i-COSTE 2023.

PROJECTS

Robust Classification Under Domain Shift: Asbestos-Cement Roofs from Satellite Data (open source, GitHub):

- Framed roof surveying as a classification problem on 8-band satellite spectra, trained in one region and tested in an unseen one: 95.2% accuracy (F1 0.955) vs 90.2% for the dataset authors' best model, with no labels from the new region.
- Engineered brightness-robust spectral features (AUC 0.984), compared Random Forest, XGBoost, SVM, and logistic regression, and fitted the decision threshold to unlabelled new-region scores with a Gaussian Mixture Model; released reproducible code and results.

Vision-Language Model Evaluation for Computer Vision Tasks (LFOVIA Lab, IIT Hyderabad):

- Evaluated VLMs as alternatives to task-specific models across object detection, segmentation, OCR, age and gender classification, crowd analysis, and action recognition, with Python benchmarking pipelines (precision/recall, IoU).
- Curated and annotated RGB and depth datasets with preprocessing, labeling, and quality-validation pipelines.

Generative AI and RAG Platform (IIT Hyderabad Build Program):

- Built a multi-tenant generative-AI platform (voice, chat, email) with a hybrid semantic + keyword RAG pipeline and LLM-based intent classification over unstructured text, improving answer relevance by 18%; trained and tested speech-to-text and text-to-speech models.

THESIS PROJECT

Machine Learning for Non-Oxidative Methane Coupling in a DBD Reactor

- Developed and benchmarked Linear Regression, Random Forest, XGBoost, and Gaussian Process Regression models to predict reactor performance from operating conditions, validated with Leave-One-Out Cross-Validation (LOOCV), reaching cross-validated R^2 up to 0.996 across six outputs from only 24 experiments; identified Bayesian Optimization for experiment-guided optimization.
- Adapted the RRSI method to statistically verify that model improvements exceed experimental noise.

TECHNICAL SKILLS

Python & ML Ecosystem: Python, SQL, C++, Linux, PyTorch, TensorFlow, Hugging Face Transformers, OpenCV, Scikit-learn, XGBoost, NumPy, Pandas, CUDA

Computer Vision & Deep Learning: Image Processing, Feature Extraction, Object Detection (YOLO), Tracking (DeepSORT), Segmentation, CNN, Vision-Language Models, Data Annotation & Augmentation

Generative AI & NLP: LLMs, Transformers, RAG, Hybrid Semantic + Keyword Retrieval, LLM Intent Classification, LLM / VLM Fine-Tuning, Speech-to-Text, Text-to-Speech

Classical ML & Statistics: Regression, Random Forest, SVM, XGBoost, Gaussian Processes, Gaussian Mixture Models, EDA, Feature Engineering, Cross-Validation, Statistical Testing, Precision / Recall / F1 / AUC / IoU

CONFERENCE PUBLICATIONS

[1] Wanekar A. D., Mannam N. P. B., Rajalakshmi P. "Novel Approach to Underwater Object Detection Using Sonar Sensors for Autonomous Underwater Vehicles (AUVs)." IEEE i-COSTE 2023, pp. 1–5.

[2] Wanekar A. D., Mannam N. P. B., Rajalakshmi P. "Object Detection and Classification for Autonomous Surface Vehicles (ASVs) Through Near-Infrared Imaging." IEEE i-COSTE 2023, pp. 1–6.

HACKATHONS, ACHIEVEMENTS & LANGUAGES

- Top 10 All India - Winner, Meta x Pragati AI Hackathon.
- AI and X Hackathon, IIT Hyderabad participant

Languages: English, Hindi, Marathi, Japanese (Basic)